Reproducibility Report for Expectation-driven sensory adaptations support enhanced acuity during categorical perception by Tim Sainburg et. al. (2025, Nature Neuroscience)

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Introduction

Tim Sainburg's 2025 paper includes a breadth of modern techniques and analysis used in the Gentner Lab, making it a useful paper to simulate in order to familiarize myself with relevant concepts and data. I will be simulating behavioral data from a two-alternative choice task (2AC) where the choices are modulated by the reliablilty of an auditory primer cue. My main

analysis goal is to find a significant difference in the population psychometric curve inflection points for decision making based on the reliability of the primer cue. This simulation will engage my interest in sensory systems while allowing me to work with real data and get experience analyzing it.

For this project, I will need to simulate a new data set, clean it to calculate the necessary analytic points (such as change in inflection point), create a linear mixed effects model, and create a 2 distribution histogram (figure 3e in the original paper). I will be aiming to replicate analysis with statistics similar to those reported in the paper ($= -11.12$, s.e.m. $= 1.158$, $z = -9.6$, $P < 0.001$) I believe that this will be an adequate challenge because it is a large dataset, and I do not have any experience with linear mixed effects models. I will need to ensure that I am cleaning and analyzing such a large dataset accurately and take the necessary steps to understand linear mixed effects models. Another challenge will be choosing simulation parameters that accuratley model the behavioral data. Although the data is not available in this paper, it is available in another paper (Gorman et al 2026) that analyzes the same dataset, making it easier to simulate an accurate representation of the data.

Learning Goals: My main goal is to refamiliarize myself with data analysis in R. I have used the language a few times for smaller projects, but without consistent use, I stil do not feel confident in my programming abilities. Although I anticipate challenges due to my inexperience, my conceptual understanding of the data, paper, and general programming knowledge make the scope of this project feasible for me. I also want to use this project as an opportunity to gain a greater understanding of linear mixed effects models, what they can tell us, and how to use them.

Links

Project repository: https://github.com/willhammond03/sainburg2026.git

Original paper: https://doi.org/10.1038/s41593-025-01899-1

Github repository for paper with accessible data: https://github.com/juliagorman/anticipatory_geometry_paper.git

Methods

Description of the steps required to reproduce the results

Please describe all the steps necessary to reproduce the key result(s) of this study.

Differences from original study

Explicitly describe known differences in the analysis pipeline between the original paper and yours (e.g., computing environment). The goal, of course, is to minimize those differences, but differences may occur. Also, note whether such differences are anticipated to influence your ability to reproduce the original results.

Project Progress Check 1

Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

Results

Data preparation

Data preparation following the analysis plan.

Key analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.